# Python Activity-Based Learning
# Program 1: University Scholarship Allocation

## Problem Statement

Evaluate scholarship applicants using aggregate marks, subject-wise minimum scores, attendance, and family income. Display eligible applicants in merit order and resolve ties using attendance.

## Objectives

• Check academic, attendance, and income eligibility.

• Calculate aggregate percentage.

• Rank eligible applicants by percentage, then attendance.

## Assumptions

• Each applicant has marks in three subjects, each out of 100.

• Minimum aggregate is 75%, each subject must be at least 60, attendance must be at least 85%, and annual family income must not exceed ■2,50,000.

• These are sample rules; replace them if your faculty provides different criteria.

## Algorithm

1. Start and read the number of applicants.

2. For each applicant, read name, three subject marks, attendance, and annual income.

3. Calculate aggregate percentage.

4. Check all eligibility conditions.

5. Store eligible applicants with percentage and attendance.

6. Sort by percentage descending, then attendance descending.

7. Display the merit list and stop.

## Flowchart

```
START → Read applicant count → Read applicant details → Calculate percentage → Check eligibility → Eligible? (
```

## Pseudocode

```
BEGIN
  READ n
  eligible = empty list
  REPEAT n times
    READ name, m1, m2, m3, attendance, income
    percentage = (m1 + m2 + m3) / 3
    IF each mark >= 60 AND percentage >= 75
       AND attendance >= 85 AND income <= 250000
       ADD (name, percentage, attendance) to eligible
    END IF
  END REPEAT
  SORT eligible by percentage descending, attendance descending
  DISPLAY eligible
END
```

## Python Source Code

```
applicants = []
n = int(input("Enter number of applicants: "))

for i in range(n):
    print("\nApplicant", i + 1)
    name = input("Enter name: ")
    m1 = int(input("Enter subject 1 marks: "))
    m2 = int(input("Enter subject 2 marks: "))
```

```
    m3 = int(input("Enter subject 3 marks: "))
    attendance = float(input("Enter attendance percentage: "))
    income = float(input("Enter annual family income: "))

    percentage = (m1 + m2 + m3) / 3

    if (m1 >= 60 and m2 >= 60 and m3 >= 60
            and percentage >= 75
            and attendance >= 85 and income <= 250000):
        applicants.append((name, percentage, attendance))

applicants.sort(key=lambda x: (x[1], x[2]), reverse=True)

print("\nSCHOLARSHIP MERIT LIST")
if len(applicants) == 0:
    print("No eligible applicants.")
else:
    for rank, student in enumerate(applicants, start=1):
        print(rank, student[0],
              "Percentage:", round(student[1], 2),
              "Attendance:", student[2])
```

## Test Case

Input: 2 applicants. Asha: 90, 85, 88; attendance 92; income 180000. Ravi: 70, 65, 72; attendance 90; income 150000. Expected: Asha appears in the merit list; Ravi is not eligible because aggregate is below 75%.

## Sample Output

```
SCHOLARSHIP MERIT LIST
1 Asha Percentage: 87.67 Attendance: 92.0
```

## Conclusion

The program checks all scholarship conditions and produces a ranked list of eligible applicants. The tie-break rule gives priority to higher attendance when percentages are equal.